Problem

The function

$$f(x) = \frac{ax+b}{cx+d},$$

where a, b, c, d are nonzero real numbers, satisfies

$$f(19) = 19, \qquad f(97) = 97,$$

and

$$f(f(x)) = x$$

for all values of x for which both sides are defined.
Determine the range of f.
Source: **AIME 1997**

§1 Exploiting the Involution Property

Since $f(f(x)) = x$, the function is its own inverse.

Substituting $f(x)$ into itself yields

$$f(f(x)) = \frac{a\left(\frac{ax+b}{cx+d}\right)+b}{c\left(\frac{ax+b}{cx+d}\right)+d} = \frac{(a^2+bc)x+b(a+d)}{c(a+d)x+(bc+d^2)}.$$

Because this expression equals x identically,

$$(a^2+bc)x+b(a+d) = x\big(c(a+d)x+(bc+d^2)\big).$$

Rearranging gives

$$c(a+d)x^2+(d^2-a^2)x-b(a+d) = 0$$

for all x.

Hence every coefficient must vanish. In particular,

$$c(a+d) = 0.$$

Since $c \neq 0$, we conclude that

$$a+d = 0,$$

or

$$d = -a.$$

§2 Using the Fixed Points

The condition $f(19) = 19$ gives

$$19 = \frac{19a + b}{19c - a}.$$

After simplification,

$$19^2 c = 2 \cdot 19a + b. \tag{1}$$

Likewise, $f(97) = 97$ gives

$$97^2 c = 2 \cdot 97a + b. \tag{2}$$

Subtracting (1) from (2),

$$(97^2 - 19^2)c = 2(97 - 19)a.$$

Factoring,

$$(97 - 19)(97 + 19)c = 2(97 - 19)a.$$

Cancelling the common factor $97 - 19$ yields

$$97 + 19 = 2\frac{a}{c}.$$

Therefore

$$a = 58c.$$

Substituting into (1),

$$361c = 2204c + b,$$

and hence

$$b = -1843c.$$

Therefore

$$f(x) = \frac{58cx - 1843c}{cx - 58c} = \frac{58x - 1843}{x - 58}.$$

§3 Revealing the Missing Value

Rewrite the function as

$$f(x) = 58 + \frac{1521}{x - 58}.$$

Since

$$1521 = 39^2 \neq 0,$$

the second term can never vanish. Consequently,

$$f(x) \neq 58$$

for every admissible x.

It remains to show that every other real number occurs.

Let $y \neq 58$. Solving

$$y = 58 + \frac{1521}{x - 58}$$

for x gives

$$x = 58 + \frac{1521}{y - 58},$$

which is a real number. Thus every real number other than 58 lies in the image of f.

§4 The Final Conclusion

The function attains every real value except 58. Therefore the range of f is

$$\boxed{\mathbb{R} \setminus \{58\}}.$$